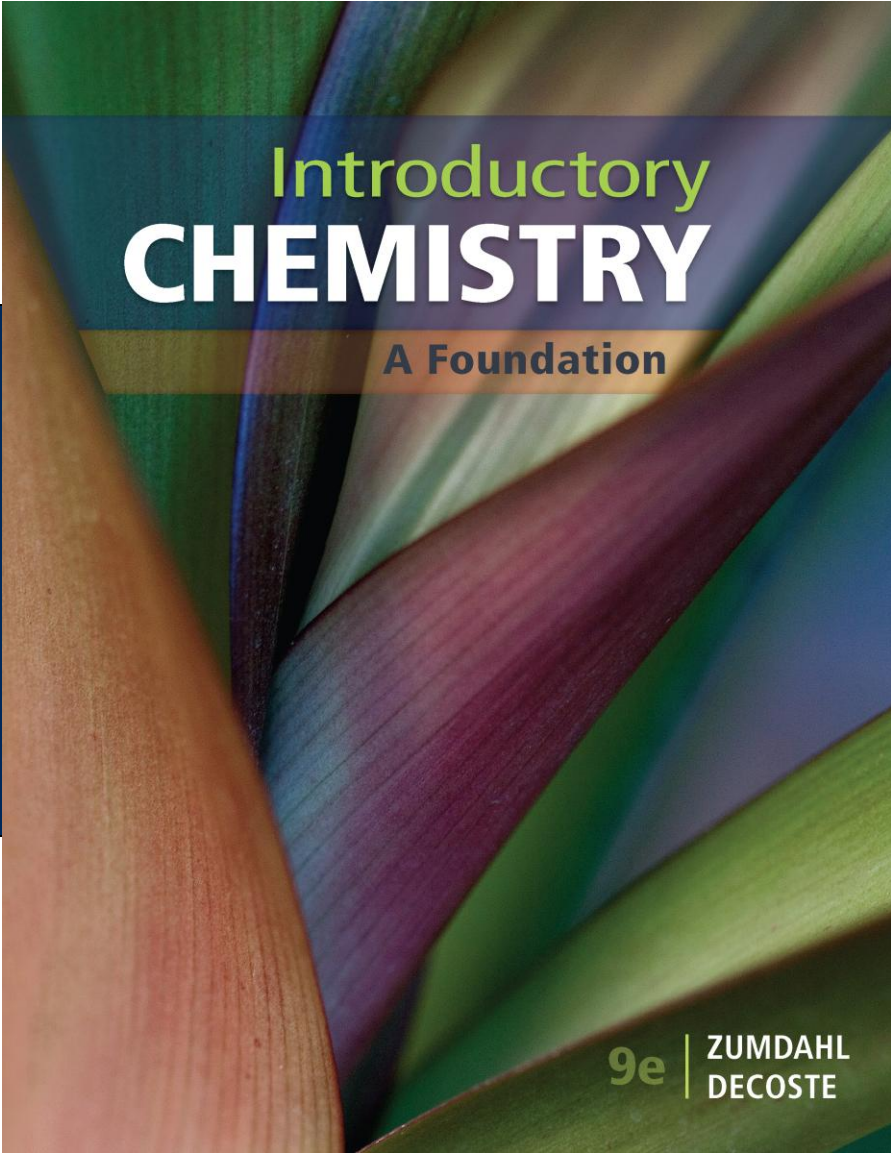




Introductory **CHEMISTRY**

A Foundation

9e | ZUMDAHL
DECOSTE

Chapter 4

Chemical Foundations: Elements, Atoms, and Ions

The Elements

Elements

- 118 known
 - 88 occur naturally
 - Rest have been made in laboratories
- Vary in abundance
 - Only 9 elements account for most of the compounds found in the Earth's crust
 - These 9 elements account for over 98% of the total mass of the Earth's crust, Oceans and Atmosphere
- List of elements found in living matter is very different from those found in the Earth's crust
 - Oxygen, carbon and nitrogen forms the basis for all biologically important molecules
 - Trace elements present in the body are very crucial

Section 4.1

The Elements

Distribution of 18 Most Abundant Elements in the Earth's Crust, Oceans, and Atmosphere

Element	Mass Percent	Element	Mass Percent
oxygen	49.2	titanium	0.58
silicon	25.7	chlorine	0.19
aluminum	7.50	phosphorus	0.11
iron	4.71	manganese	0.09
calcium	3.39	carbon	0.08
sodium	2.63	sulfur	0.06
potassium	2.40	barium	0.04
magnesium	1.93	nitrogen	0.03
hydrogen	0.87	fluorine	0.03
		all others	0.49

The Elements

How the Term Element is Used

- Could mean a single atom of an element (Ar or H)
- Could mean molecules of an element (H_2), which is hydrogen found in its natural state
- Could mean atoms of an element are present in some form (sodium found in the human body)
- Look at each particular case to determine its proper use

Symbols for the Elements

Element Names and Symbols

- Each element has a unique one- or two-letter symbol
- First letter is always capitalized and the second is not
- The symbol usually consists of the first or first two letters of the element's name
 - Oxygen: O
 - Krypton: Kr
- Sometimes the symbol is taken from the element's original Latin or Greek name
 - Gold: Au from aurum
 - Lead: Pb from plumbum

Section 4.2

Symbols for the Elements

Names and Symbols of the Most Common Elements

Element	Symbol	Element	Symbol
aluminum	Al	lithium	Li
antimony (stibium)*	Sb	magnesium	Mg
argon	Ar	manganese	Mn
arsenic	As	mercury (hydrargyrum)	Hg
barium	Ba	neon	Ne
bismuth	Bi	nickel	Ni
boron	B	nitrogen	N
bromine	Br	oxygen	O
cadmium	Cd	phosphorus	P
calcium	Ca	platinum	Pt
carbon	C	potassium (kalium)	K
chlorine	Cl	radium	Ra
chromium	Cr	silicon	Si
cobalt	Co	silver (argentum)	Ag
copper (cuprum)	Cu	sodium (natrium)	Na
fluorine	F	strontium	Sr
gold (aurum)	Au	sulfur	S
helium	He	tin (stannum)	Sn
hydrogen	H	titanium	Ti
iodine	I	tungsten (wolfram)	W
iron (ferrum)	Fe	uranium	U
lead (plumbum)	Pb	zinc	Zn

*Where appropriate, the original name is shown in parentheses so that you can see where some of the symbols came from.

Dalton's Atomic Theory

Atomic composition of Elements and Compounds

- Most natural materials are mixtures of pure substances
- Pure substances are either elements or combinations of elements called compounds
- A given compound always contains the same proportions, by mass, of the elements

Dalton's Atomic Theory

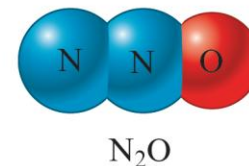
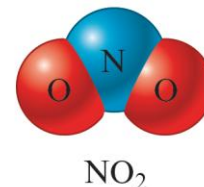
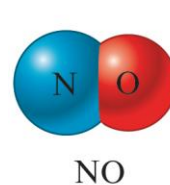
Law of Constant Composition

- A given compound always has the same composition, regardless of where it comes from
 - Water always contains 8 g of oxygen for every 1 g of hydrogen
 - Carbon dioxide always contains 2.7 g of oxygen for every 1 g of carbon

Dalton's Atomic Theory

Dalton's Atomic Theory

- Elements are made of tiny particles called atoms
- All atoms of a given element are identical
- The atoms of a given element are different from those of any other element
- Atoms of one element can combine with atoms of other elements to form compounds
 - A given compound always has the same relative numbers and types of atoms



Dalton's Atomic Theory

Dalton's Atomic Theory (continued)

- Atoms are indivisible in chemical processes
 - Atoms are not created or destroyed in chemical reactions
 - A chemical reaction simply changes the way the atoms are grouped together

Dalton's Atomic Theory

Concept Check

Which of the following statements regarding Dalton's atomic theory are still believed to be true?

- Elements are made of tiny particles called atoms
- All atoms of a given element are identical
- A given compound always has the same relative numbers and types of atoms
- Atoms are indestructible

Formulas of Compounds

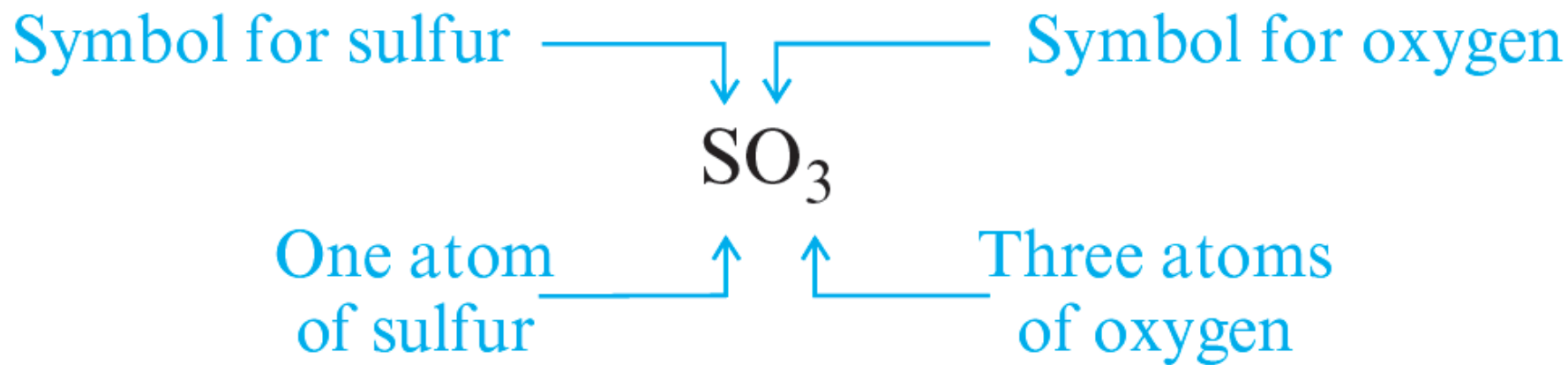
Chemical Formulas Describe Compounds

- Compound: Distinct substance that is composed of atoms of two or more elements and always contains exactly the same relative masses of those elements
- Chemical Formulas: Express the types of atoms and the number of each type in each unit, or molecule, of a given compound

Formulas of Compounds

Rules for Writing Formulas

- Each atom present is represented by its element symbol
- The number of each type of atom is indicated by a subscript written to the right of the element symbol
- When only one atom of a given type is present, the subscript 1 is not written



Formulas of Compounds

Formulas of Compounds: Exercise

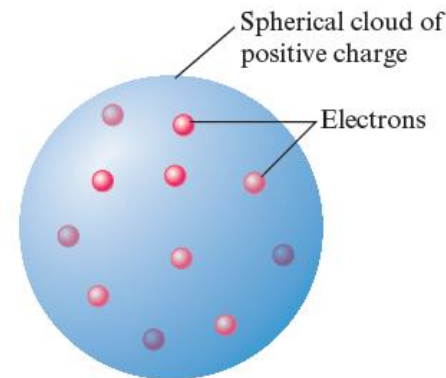
The pesticide known as DDT paralyzes insects by binding to their nerve cells, leading to uncontrolled firing of the nerves. Before most of the uses of DDT were banned in the U.S., many insects had developed a resistance to it. Write out the formula for DDT. It contains 14 carbon atoms, 9 hydrogen atoms, and 5 atoms of chlorine



The Structure of the Atom

William Thomson (Plum Pudding Model)

- Reasoned that the atom might be thought of as a uniform “pudding” of positive charge with enough negative electrons scattered within to counterbalance that positive charge



Plum pudding model of the atom.



English plum pudding.

Photo by bit245/Depositphotos.com

The Structure of the Atom

Ernest Rutherford (1911)

- Conducted experiment on alpha particle bombardment of metal foil
 - Concluded from the results that the plum pudding model is wrong
 - Atom has a dense center of positive charge called the nucleus
 - Electrons travel around the nucleus at a relatively large distance
 - A proton has the same magnitude of charge as the electron, but its charge is positive

The Structure of the Atom

Rutherford and Chadwick (1932)

- Most nuclei also contain a neutral particle called the neutron
- A neutron is slightly more massive than a proton but has no charge

Introduction to the Modern Concept of Atomic Structure

The atom contains

- Electrons: Negatively charged and found outside the nucleus
- Protons: Found in the nucleus with positive charge, equal in magnitude to the electron's negative charge
- Neutrons: Found in the nucleus with no charge and virtually same mass as a proton

Section 4.6

Introduction to the Modern Concept of Atomic Structure

Mass and Charge of the Electron, Proton, and Neutron

Particle	Relative Mass*	Relative Charge
electron	1	1—
proton	1836	1+
neutron	1839	none

*The electron is arbitrarily assigned a mass of 1 for comparison.

Introduction to the Modern Concept of Atomic Structure

Why do different atoms have different chemical properties?

- The chemistry of an atom arises from its electrons
- Electrons are the parts of atoms that “intermingle” when atoms combine to form molecules
- It is the number of electrons that really determines chemical behavior

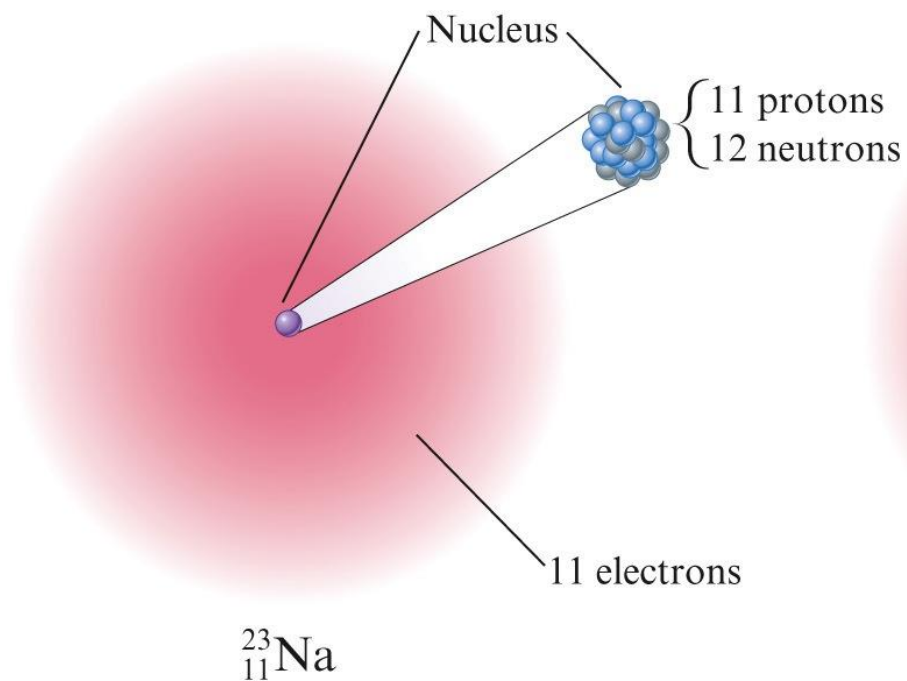
Isotopes

Isotopes

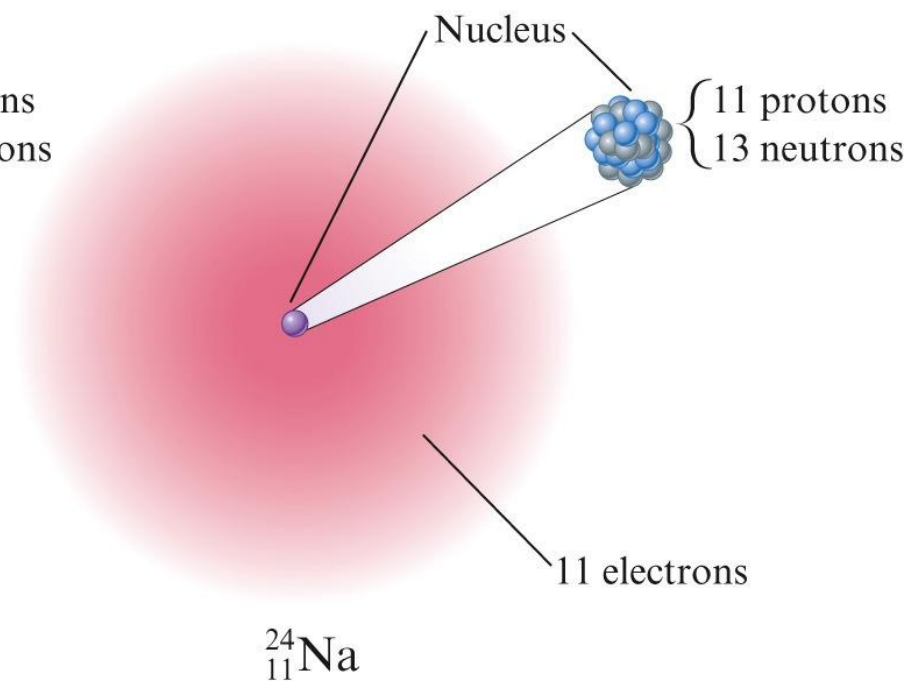
- Atoms with the same number of protons but different numbers of neutrons
- Show almost identical chemical properties
 - Chemistry of an atom is due to its electrons
- In nature, elements are usually found as a mixture of isotopes

Isotopes

Two Isotopes of Sodium



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Isotopes

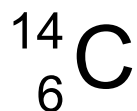
Isotopes (continued)



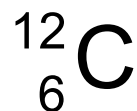
- X = the symbol of the element
- A = the mass number
 - Number of protons and neutrons
- Z = the atomic number
 - Number of protons

Isotopes

Isotopes: An Example



- C = the symbol for carbon
- 6 = the atomic number
 - 6 protons
- 14 = the mass number
 - 6 protons and 8 neutrons



- C = the symbol for carbon
- 6 = the atomic number
 - 6 protons
- 12 = the mass number
 - 6 protons and 6 neutrons

Isotopes: Exercise

A certain isotope X contains 23 protons and 28 neutrons

- What is the mass number of this isotope?
- Identify the element

Mass Number = 51

Vanadium

Introduction to the Periodic Table

The Periodic Table (continued 1)

- Metals vs. Nonmetals
- Groups or Families: Elements in the same vertical columns and have similar chemical properties
- Periods: Horizontal rows of elements

Introduction to the Periodic Table

The Periodic Table (continued 2)

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- Most elements are metals and occur on the left side
- The nonmetals appear on the right side
- Metalloids are elements that have some metallic and some nonmetallic properties

Introduction to the Periodic Table

Physical Properties of Metals

- Efficient conduction of heat and electricity
- Malleability
 - They can be hammered into thin sheets
- Ductility
 - They can be pulled into wires
- Lustrous appearance

Introduction to the Periodic Table

Physical Properties of Nonmetals

- Lack properties of metals
- Exhibit more variation in properties
- Can be gases, liquids, or solids at room temperature

Introduction to the Periodic Table

Physical Properties of Metalloids

- Exhibit a mixture of metallic and non-metallic properties

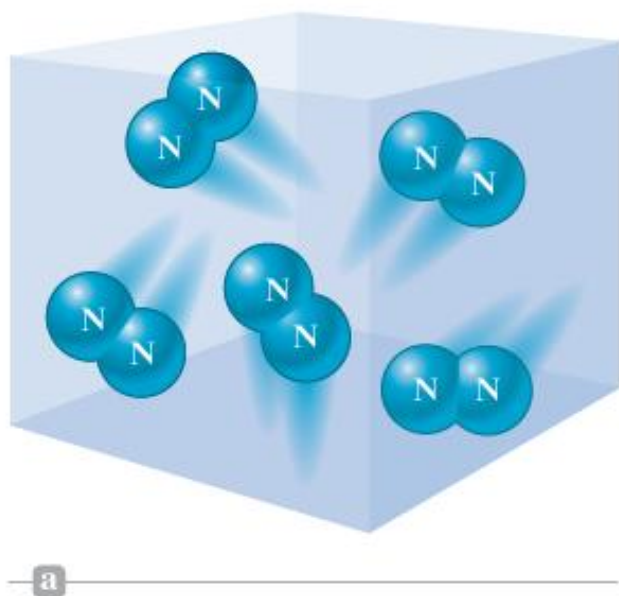
Natural States of the Elements

Natural States of the Elements

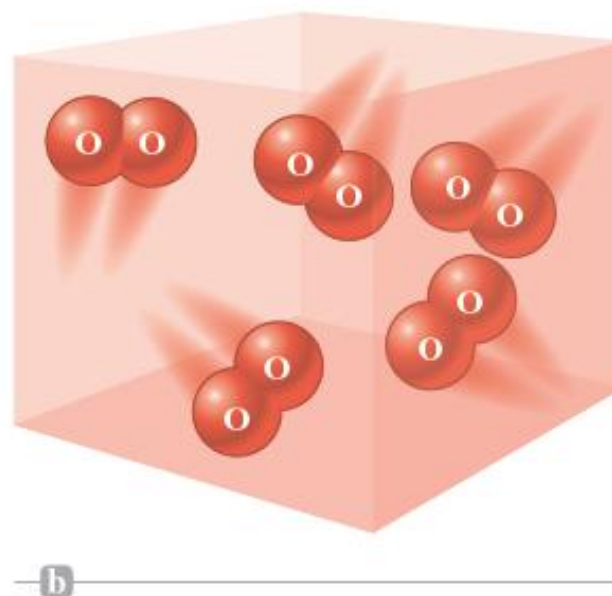
- Most elements are quite reactive
- Elements are not generally found in pure form
 - Exceptions
 - Noble metals: Gold, platinum and silver
 - Noble gases: Group 8

Natural States of the Elements

Diatomic Molecules



- Nitrogen gas contains N_2 molecules



- Oxygen gas contains O_2 molecules

Natural States of the Elements

Elements That Exist as Diatomic Molecules in Their Elemental Forms

Element Present	Elemental State at 25 °C	Molecule
hydrogen	colorless gas	H ₂
nitrogen	colorless gas	N ₂
oxygen	pale blue gas	O ₂
fluorine	pale yellow gas	F ₂
chlorine	pale green gas	Cl ₂
bromine	reddish-brown liquid	Br ₂
iodine	lustrous, dark purple solid	I ₂

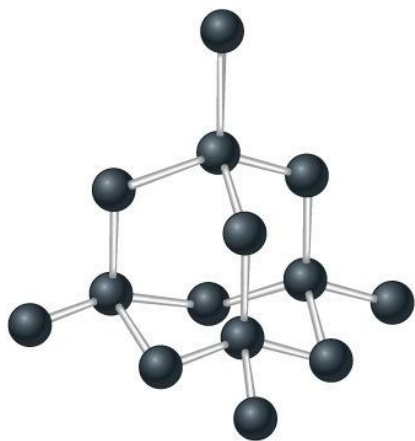
Natural States of the Elements

Allotropes

- Different forms of a given element
- Example
 - Solid carbon occurs in three forms
 - Diamond
 - Graphite
 - Buckminsterfullerene

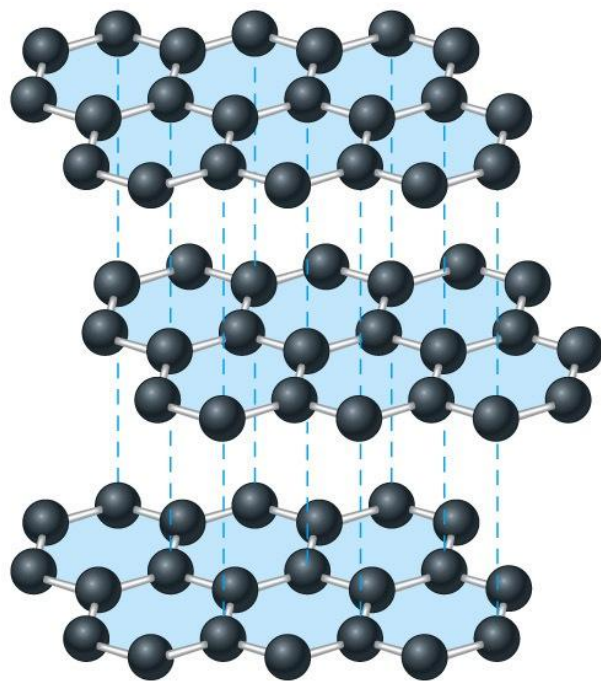
Natural States of the Elements

Allotropes of Carbon

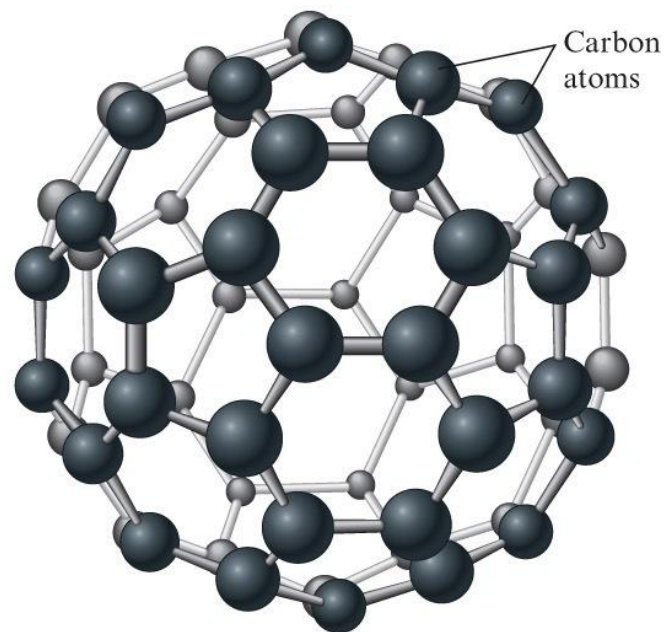


Diamond

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Graphite

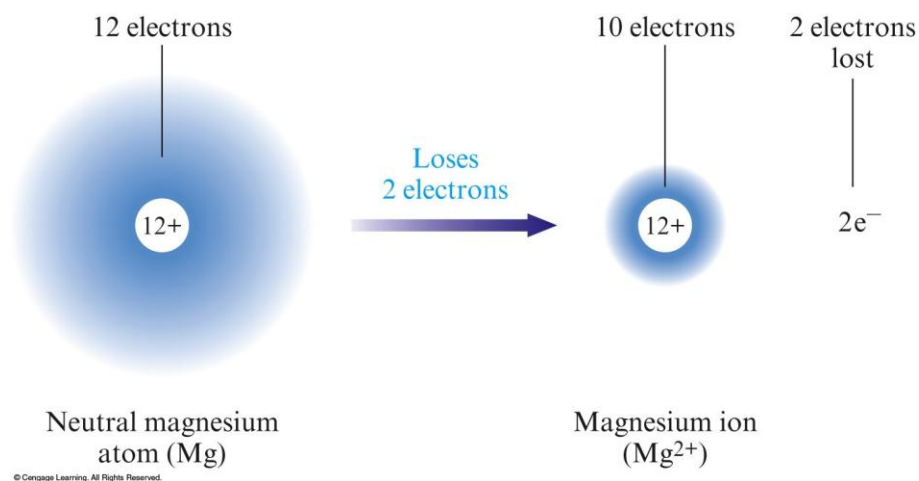


Buckminsterfullerene

Ions

Formation of Ions

- Atoms can form ions by gaining or losing electrons
 - Metals tend to lose one or more electrons to form positive ions called cations

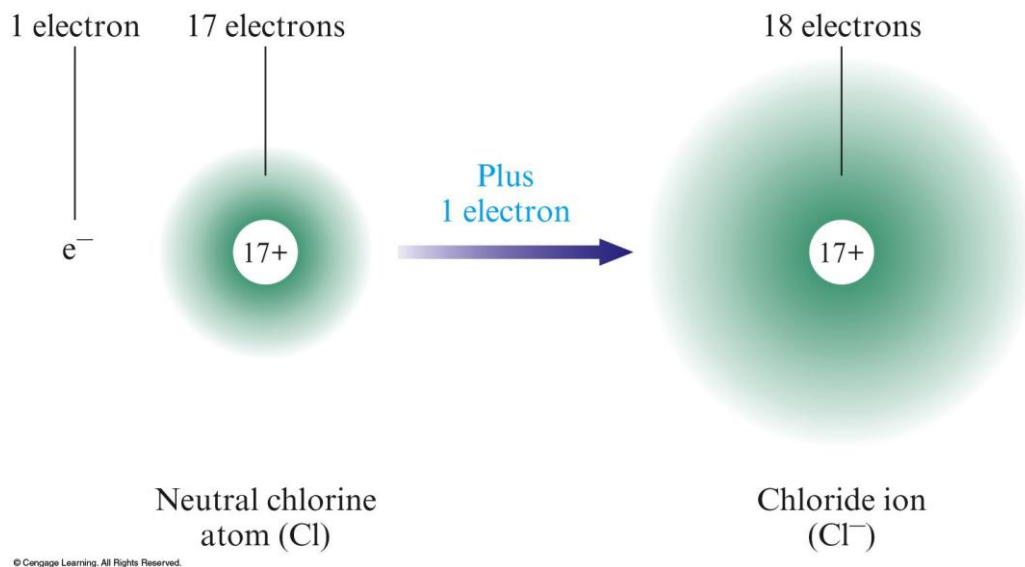


- Cations are generally named by using the name of the parent atom

Ions

Anions

- Nonmetals tend to gain one or more electrons to form negative ions called anions



- Anions are named by using the root of the atom name followed by the suffix **-ide**

Ions

Ion Charges and the Periodic Table

- The ion that a particular atom will form can be predicted from the atom's position on the periodic table

Group or Family	Charge
Alkali Metals (1A)	1+
Alkaline Earth Metals (2A)	2+
Halogens (7A)	1–
Noble Gases (8A)	0

Ions

Ion Charges and the Periodic Table (continued)

1	2							3	4	5	6	7	8
											O ²⁻	F ⁻	
Li ⁺	Be ²⁺							Al ³⁺			S ²⁻	Cl ⁻	
Na ⁺	Mg ²⁺							Ga ³⁺			Se ²⁻	Br ⁻	
K ⁺	Ca ²⁺							In ³⁺			Te ²⁻	I ⁻	
Rb ⁺	Sr ²⁺												
Cs ⁺	Ba ²⁺												

Transition metals form cations
with various charges.

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Ions

Exercise 1

An ion with a 3+ charge contains 23 electrons. Which ion is it?

- a) Fe^{3+}
- b) V^{3+}
- c) Ca^{3+}
- d) Sc^{3+}

Exercise 2

A certain ion X^+ contains 54 electrons and 78 neutrons

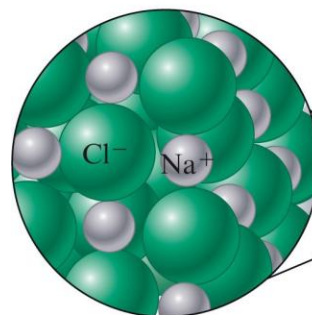
- What is the **mass number** of this ion?

133

Compounds That Contain Ions

Ionic Compounds

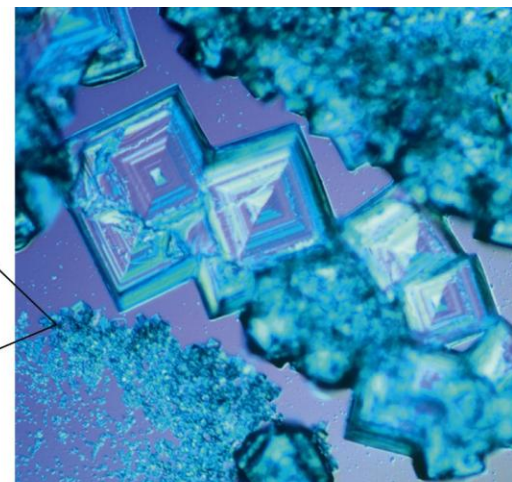
- Ions combine to form ionic compounds
- Properties of ionic compounds
 - High melting points
 - Conduct electricity
 - If melted
 - If dissolved in water



a

The arrangement of sodium ions (Na⁺) and chloride ions (Cl⁻) in the ionic compound sodium chloride.

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b

Solid sodium chloride highly magnified.

SMC Images/Getty Images

Compounds That Contain Ions

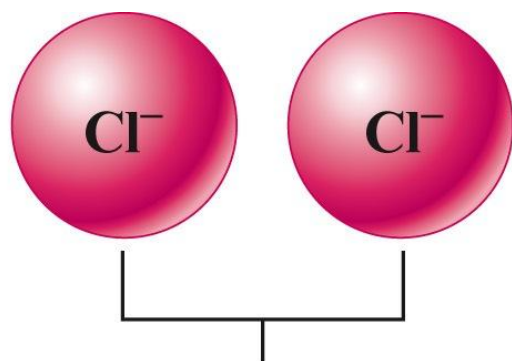
Ionic Compounds (continued)

- Ionic compounds are electrically neutral
- The charges on the anions and cations in the compound must sum to zero

Compounds That Contain Ions

Formulas for Ionic Compounds

- Write the cation element symbol followed by the anion element symbol
- The number of cations and anions must be correct for their charges to sum to zero



Cation charge:
 $2+$

+

Anion charge:
 $2 \times (1-)$

=

Compound net
charge: 0

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Compounds That Contain Ions

Concept Check 1

A compound contains an unknown ion X and has the formula XCl_2 . Ion X contains 20 electrons. What is the identity of X?

- a) Ti^{2+}
- b) Sc^+
- c) Ca^{2+}
- d) Cr^{2+}

Compounds That Contain Ions

Concept Check 2

A member of the alkaline earth metal family whose most stable ion contains 36 electrons forms a compound with bromine. What is the **correct formula** for this compound?

- a) CaBr_2
- b) KrBr
- c) RbBr
- d) SrBr_2